

# Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo + Discrepancy

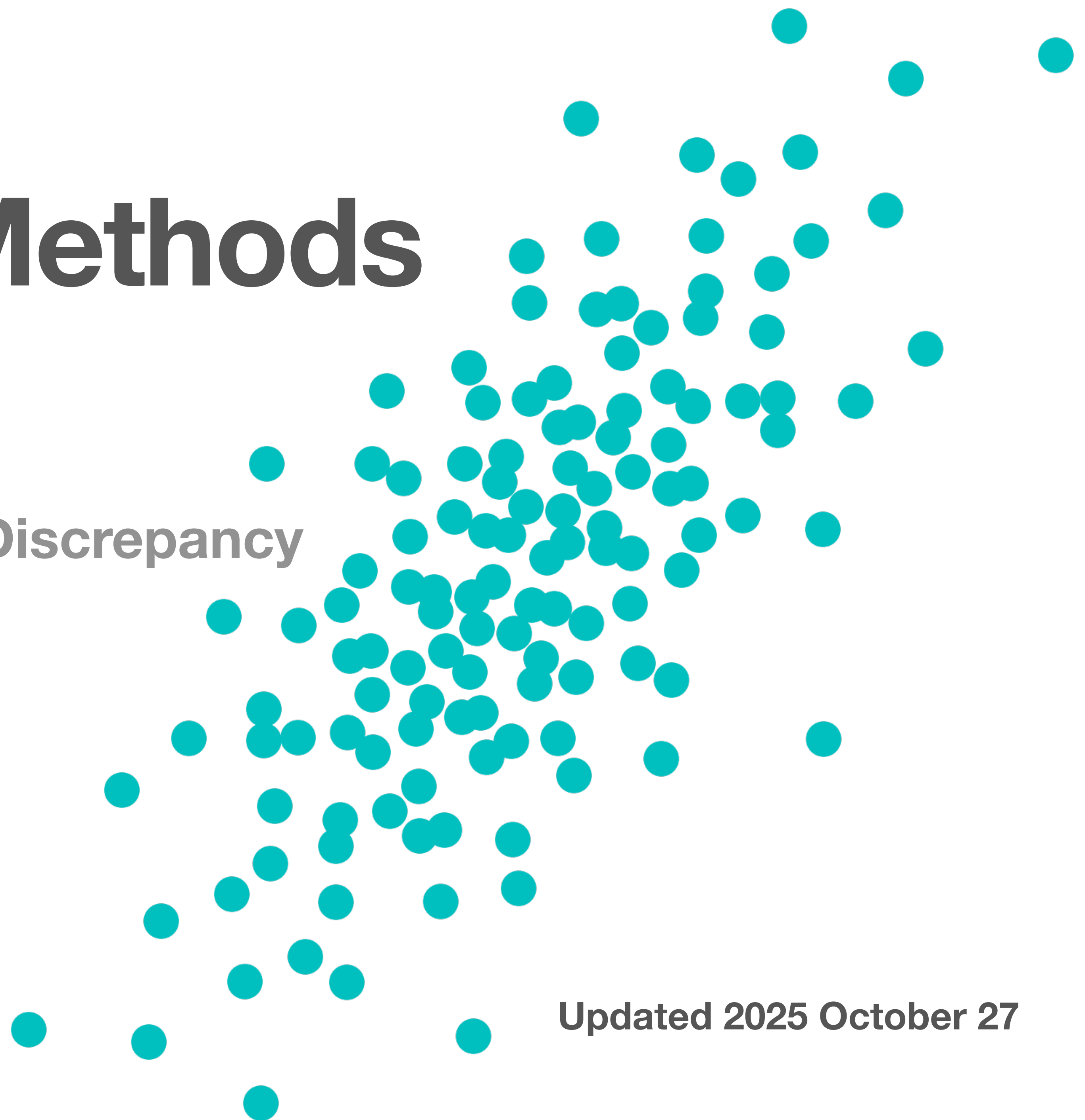
Improving Efficiency

Selected Topics

Git website and repository  
Canvas

Fred Hickernell, Fall 2025

Updated 2025 October 27

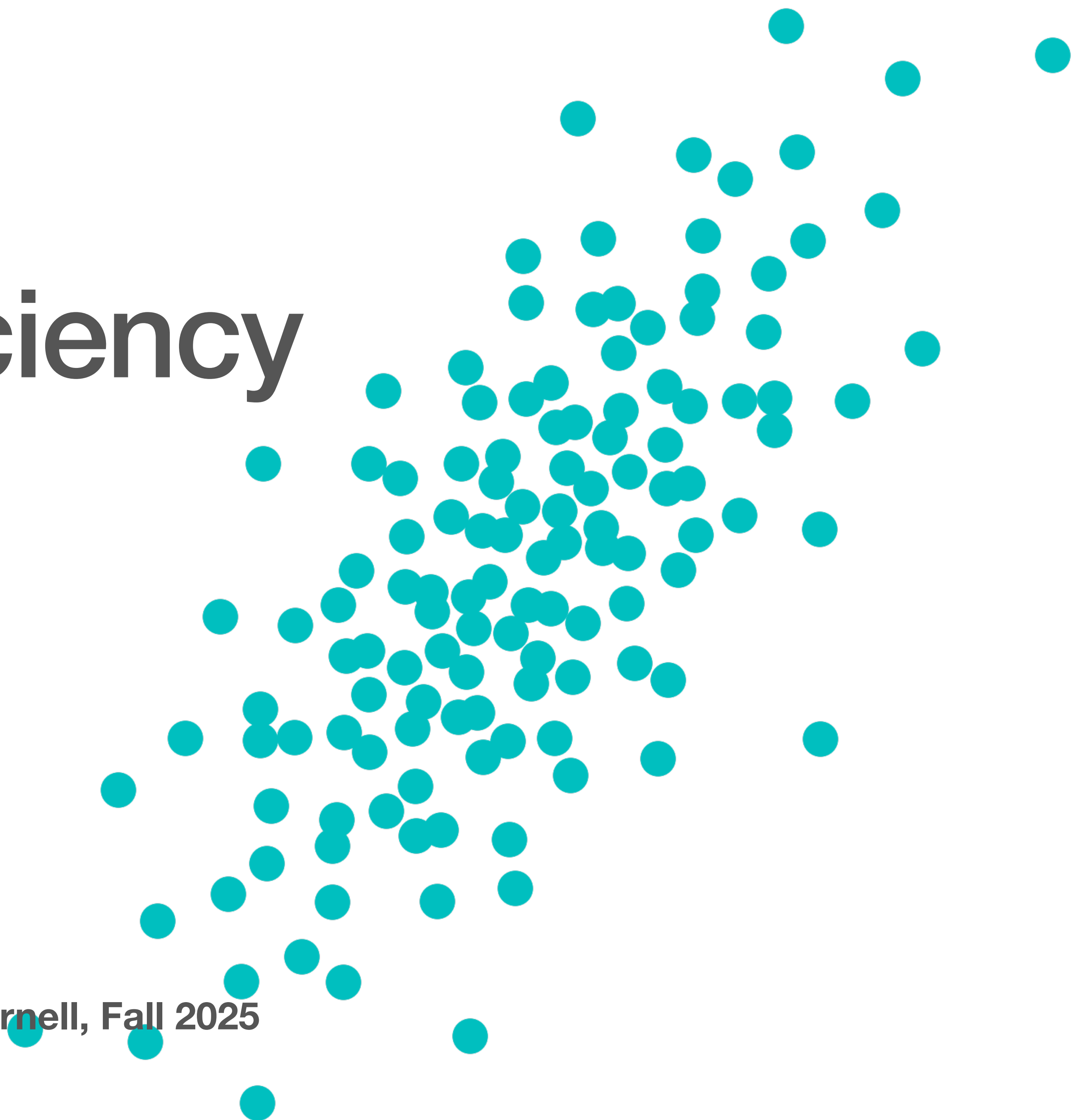


# Improving Efficiency

Owen, Chapters 8–10, 15–17

Test 2 Oct 29

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025





# Variance Reduction

$$\mu = \mathbb{E}[f(X)] = \mathbb{E}[g(X)], \quad X \sim \mathcal{U}[0,1]^d$$

Is it better to use

$$\hat{\mu}_{f,n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \quad \text{or} \quad \hat{\mu}_{g,n} = \frac{1}{n} \sum_{i=0}^{n-1} g(X_i), \quad X_0, X_1, \dots \stackrel{\text{iid}}{\sim} \mathcal{U}[0,1]^d$$

to approximate  $\mu$ ? Since both sample means are unbiased, we can compare their variances (mean squared errors):

$$\text{var}(\hat{\mu}_{f,n}) = \frac{\text{var}(f(X))}{n} \quad \text{var}(\hat{\mu}_{g,n}) = \frac{\text{var}(g(X))}{n}$$

so it depends on the variances of the two functions





# Keister example

$$\mu = \int_{\mathbb{R}^d} \cos(\|t\|) \exp(-\|t\|^2) dt$$

$$\mathbf{x} = (\Phi(at_1), \dots, \Phi(at_d)), \quad \Phi \text{ is the CDF of the standard normal}$$

$$d\mathbf{x} = a^d \phi(at_1) \cdots \phi(at_d) dt = \frac{a^d \exp(-a^2 \|t\|^2/2)}{(2\pi)^{d/2}} dt$$

$$t = (\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d)) / a$$

$$\mu = \int_{[0,1]^d} \frac{(2\pi)^{d/2}}{a^d} \cos\left(\left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\| / a\right)$$

$$\times \exp\left(-\frac{2-a^2}{2a^2} \left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\|^2\right) d\mathbf{x}, \quad (a^2 \leq 2)$$

Whole family  $f(\mathbf{x}; a)$  for which  $\mu = \mathbb{E}[f(X; a)]$  for  $X \sim \mathcal{U}[0,1]^d$



# Variable transformations

Suppose that you want to approximate

$$\mu = \int_{\mathcal{T}} g(\mathbf{t}) \, d\mathbf{t}$$

by Monte Carlo. You need to write it as an expectation. Let  $\mathbf{t} = \boldsymbol{\psi}(\mathbf{x})$ , for some **invertible transformation**,  $\boldsymbol{\psi} : \mathcal{X} \rightarrow \mathcal{T}$ . Define the **Jacobian determinant** as

$$J(\mathbf{x}; \boldsymbol{\psi}) := \det \left( \left( \frac{\partial \psi_i}{\partial x_j} \right)_{i,j=1}^d \right), \quad J(\mathbf{t}; \boldsymbol{\psi}^{-1}) := \det \left( \left( \frac{\partial \psi_i^{-1}}{\partial t_j} \right)_{i,j=1}^d \right), \quad J(\boldsymbol{\psi}(\mathbf{x}); \boldsymbol{\psi}^{-1}) = \frac{1}{J(\mathbf{x}; \boldsymbol{\psi})}$$

Then

$$\mu = \int_{\mathcal{X}} g(\boldsymbol{\psi}(\mathbf{x})) |J(\mathbf{x}; \boldsymbol{\psi})| \, d\mathbf{x} = \int_{\mathcal{X}} \underbrace{\frac{g(\boldsymbol{\psi}(\mathbf{x}))}{\varrho_{\mathbf{X}}(\mathbf{x})} |J(\mathbf{x}; \boldsymbol{\psi})|}_{f(\mathbf{x})} \varrho_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$



# Variable transformations

$$\mu = \int_{\mathcal{T}} g(t) dt = ?$$

Let  $t = \psi(x)$ , for some **invertible transformation**,  $\psi : \mathcal{X} \rightarrow \mathcal{T}$ . Then

$$\mu = \int_{\mathcal{X}} g(\psi(x)) |J(x; \psi)| dx = \int_{\mathcal{X}} \underbrace{\frac{g(\psi(x))}{\varrho(x)} |J(x; \psi)|}_{f(x)} \varrho(x) dx = \mathbb{E}_{\mathbf{X} \sim \varrho}[f(\mathbf{X})]$$

If  $\mathbf{X} \sim \mathcal{U}[0,1]^d$  (because we know how generate **uniform** samples), then

$$\mu = \int_{[0,1]^d} \underbrace{g(\psi(x)) |J(x; \psi)|}_{f(x)} dx = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d}[f(\mathbf{X})]$$





# Random variable transformations

Let

- $T$  be a random variable with sample space  $\mathcal{T}$  and density  $\varrho_T$
- $X$  be a random variable with sample space  $\mathcal{X}$  and density  $\varrho_X$
- $T = \psi(X)$ , for some **invertible transformation**,  $\psi : \mathcal{X} \rightarrow \mathcal{T}$ . Then

$$\varrho_T(t) = \varrho_X(\psi^{-1}(t)) |J(t; \psi^{-1})|, \quad \varrho_T(\psi(x)) = \varrho_X(x) |J(\psi(x)); \psi^{-1})| = \frac{\varrho_X(x)}{|J(x; \psi)|}$$

$$\mu = \int_{\mathcal{T}} \underbrace{h(t) \varrho_T(t)}_{g(t)} dt = \mathbb{E}_{T \sim \varrho_T}[h(T)] = \mathbb{E}_{X \sim \varrho_X}[h(\psi(X))] = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$

$$\text{or } = \int_{\mathcal{X}} \frac{h(\psi(x)) \varrho_T(\psi(x))}{\varrho_X(x)} |J(x; \psi)| \varrho_X(x) dx = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$



# Importance sampling

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \int_{\mathcal{T}} \underbrace{h(\mathbf{t}) \frac{\varrho_{\mathbf{T}}(\mathbf{t})}{\varrho_{\mathbf{X}}(\mathbf{t})}}_{f(\mathbf{t})} \overbrace{\varrho_{\mathbf{X}}(\mathbf{t})}^{\text{likelihood ratio}} \, d\mathbf{t} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$

We try to choose  $\varrho_{\mathbf{X}}$  to make  $\text{var}[f(\mathbf{X})]$  small when doing IID sampling. If  $\mathbf{T} = \boldsymbol{\psi}(\mathbf{X})$  for  $\mathbf{X} \sim \mathcal{X}[0,1]^d$ —since often find generate non-uniform samples from uniform ones—then

$$\mu = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d} [h(\boldsymbol{\psi}(\mathbf{X}))] = \int_{[0,1]^d} h(\boldsymbol{\psi}(\mathbf{x})) \, d\mathbf{x}$$





# Ex. Brownian motion with drift

In computational finance certain outcomes, such as positive option payoffs, may be **rare**

- Adding a drift to the Brownian motion **increases** the probability that asset paths yield positive payoffs and
- **Reduces the variance** of our estimator of the option price (population mean)
- Paths with positive payoffs receive **smaller sample weights** to compensate and keep the mean the same

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}, \quad \varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left( \min(t_i, t_j) \right)_{i,j=1}^d$$

Here the density  $\varrho$  is a discretized **Brownian motion**



# Ex. Brownian motion with drift cont'd

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}$$
$$\varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left( \min(t_i, t_j) \right)_{i,j=1}^d$$

We importance sample:

$$\varrho_{\text{drift}}(\mathbf{x}) = \frac{\exp(-(\mathbf{x} - \mathbf{a})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{a}) / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \mathbf{a} = \alpha(t_1, \dots, t_d)^\top$$

$$\text{likelihood ratio} = \frac{\varrho(\mathbf{x})}{\varrho_{\text{drift}}(\mathbf{x})} = \exp(-\alpha x_d + \alpha^2 t_d / 2)$$

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \exp(-\alpha x_d + \alpha^2 t_d / 2) \varrho_{\text{drift}}(\mathbf{x}) d\mathbf{x}$$

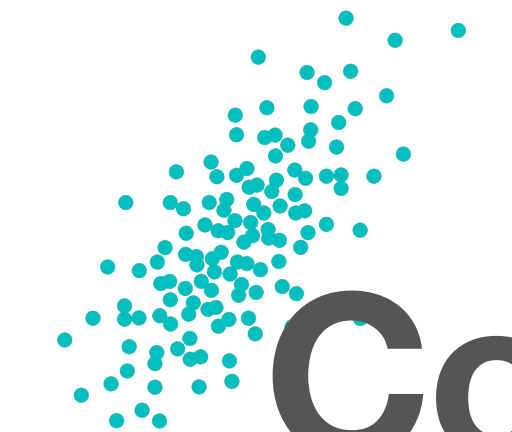
Note that the weight gets **smaller** as  $\alpha x_d$  increases or  $\alpha^2 t_d$  decreases



# Variable transformation / importance sampling observations

- Variable transformation and importance sampling are **intertwined**
- The **choice** of sampling density or variable transformation affects the variance of the sample mean
- You must choose the variable transformation so that
  - The Jacobian determinant,  $J(\mathbf{x}; \boldsymbol{\psi})$ , is **finite**
  - The probability density,  $q_{\mathbf{X}}(\mathbf{x})$ , does not **vanish** unless integrand vanishes
- You might try **pilot** samples to help choose transformation





# Control variates

$$\mu = \mathbb{E}[f(X)] \iff \mu = \mathbb{E}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)]$$

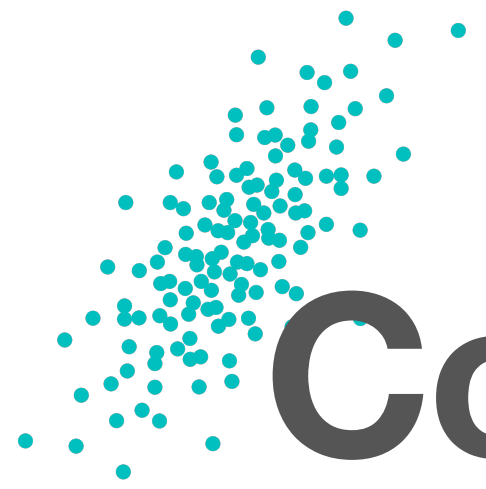
provided that  $\mathbb{E}[\eta_1(X)] = \dots \mathbb{E}[\eta_m(X)] = 0$ . Want to arrange such that

$$\text{var}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)] \leq \text{var}[f(X)]$$

Then

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \dots - \beta_m \eta_m(X_i)] \text{ converges faster than } \hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$

How do we **minimize**  $\text{var}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)]$ ?



# Control variates (cont'd)

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \cdots - \beta_m \eta_m(X_i)]$$

How do we **minimize**  $\text{var}[f(X) - \beta_1 \eta_1(X) - \cdots - \beta_m \eta_m(X)]$ ? Use least squares regression with centered response and explanatory variables to choose  $\hat{\beta}$ :

$$\hat{\beta} = \underset{\mathbf{b}}{\text{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2$$

$$\text{where } \mathbf{y} = \begin{pmatrix} f(\mathbf{x}_0) - \hat{\mu}_n \\ \vdots \\ f(\mathbf{x}_{n-1}) - \hat{\mu}_n \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} \eta_1(\mathbf{x}_0) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_0) - \hat{\mu}_{\eta_m,n} \\ \vdots & & \vdots \\ \eta_1(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_m,n} \end{pmatrix}$$

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i), \quad \hat{\mu}_{\eta_1,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_1(\mathbf{x}_i), \quad \dots, \quad \hat{\mu}_{\eta_m,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_m(\mathbf{x}_i)$$



# Least Squares

How do we solve

$$\hat{\beta} = \underset{\mathbf{b}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 = \underset{\mathbf{b}}{\operatorname{argmin}} (\mathbf{y} - \mathbf{H}\mathbf{b})^T (\mathbf{y} - \mathbf{H}\mathbf{b})$$

Let

$$\mathbf{b} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}$$

Then

$$\begin{aligned} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 &= \|\mathbf{y} - \mathbf{H}[(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}]\|^2 \\ &= \|[ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{H}\mathbf{c}\|^2 \\ &= ([ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{H}\mathbf{c})^T ([ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{H}\mathbf{c}) \\ &= \mathbf{y}^T [ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{c}^T \mathbf{H}^T \mathbf{H} \mathbf{c} \end{aligned}$$

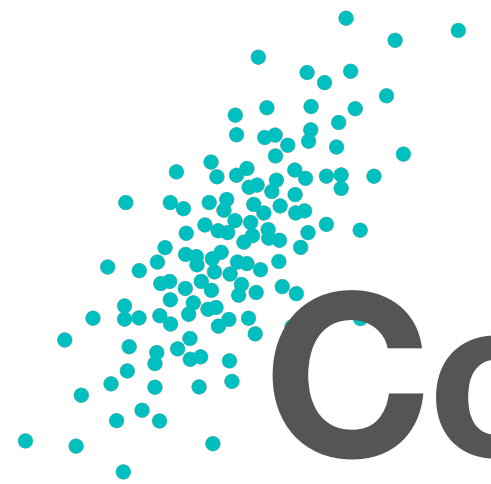
So the optimum comes by setting  $\mathbf{c} = \mathbf{0}$





# Control variates

- Should be **clever** in choosing the  $\eta_j$ 
  - **Good** choices can decrease the sample size required
  - **Bad** choices only hurt in terms of increased **computation** time, but not increased sample size
- The  $\beta_j$  should be chosen by **least squares** for IID sampling
- For **low discrepancy** sequences, the  $\beta_j$  should be chosen to reduce the **variation** (not variance)



# Conditional Monte Carlo

Recall that

$$\mathbb{E}(Y) = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})], \quad \text{var}(Y) = \mathbb{E}_{\mathbf{Z}}[\text{var}(Y | \mathbf{Z})] + \text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})]$$

Which implies that

$$\text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})] \leq \text{var}(Y), \quad \text{var} \left( \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_Y(Y | \mathbf{Z}_i) \right) \leq \text{var} \left( \frac{1}{n} \sum_{i=0}^{n-1} Y_i \right)$$

Let  $(Y = f(\mathbf{X}_{1:d}), \mathbf{X}_{1:d})$  be a vector of random variables

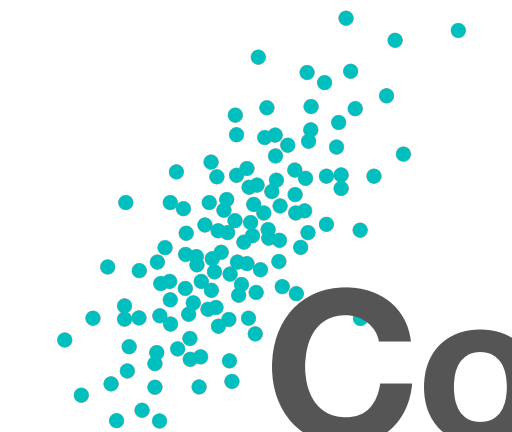
Let  $\mathbf{Z} = \mathbf{X}_{2:d}$

If  $g(\mathbf{X}_{2:d}) = \mathbb{E}_Y(Y | \mathbf{X}_{2:d})$  has an analytic form, then

$$\mu = \mathbb{E}(Y) = \mathbb{E}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})] \text{ and } \text{var}(Y) \geq \text{var}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})]$$

If  $h(y, \mathbf{X}_{2:d}) = \varrho_{Y|\mathbf{X}_{2:d}}(y | \mathbf{x}_{2:d})$  has an analytic form, then

$$\varrho_Y(y) = \mathbb{E}_{\mathbf{X}_{2:d}}[h(y, \mathbf{X}_{2:d})] \quad \text{probability density expressed as expectation}$$



# Conditional Monte Carlo for Density Estimation

Estimate the probability density,  $\varrho$ , for  $Y = f(X)$ , where  $X \sim \mathcal{U}[0,1]^d$

If one can identify an **increasing**  $g$ , such that

$$y = f(\mathbf{x}) \iff x_1 = g(y; \mathbf{x}_{2:d})$$

Then

$$\begin{aligned} F_{Y|X_{2:d}}(y | \mathbf{x}_{2:d}) &= \mathbb{P}[Y \leq y | X_{2:d} = \mathbf{x}_{2:d}] = \mathbb{P}[f(X) \leq y | X_{2:d} = \mathbf{x}_{2:d}] \\ &= \mathbb{P}[X_1 \leq g(y; \mathbf{x}_{2:d})] = g(y; \mathbf{x}_{2:d}) \end{aligned}$$

$$\varrho(y) = \mathbb{E}[\varrho_{Y|X_{2:d}}(y | \mathbf{x}_{2:d})] = \int_{[0,1]^{d-1}} g'(y; \mathbf{x}) \, d\mathbf{x}$$





## Ex. Density of a sum of uniforms

Let  $Y = \gamma_1 X_1 + \cdots + \gamma_d X_d$ , where  $X \sim \mathcal{U}[0,1]^d$ . What is the PDF of  $Y$ ?

$$g(y; x_2, \dots, x_d) = \frac{1}{\gamma_1} (y - \gamma_2 x_2 + \cdots \gamma_d x_d), \quad \gamma_2 x_2 + \cdots \gamma_d x_d \leq y \leq \gamma_1 x_1 + \cdots + \gamma_d x_d$$

$$g'(y; x_2, \dots, x_d) = \frac{1}{\gamma_1}, \quad \gamma_2 x_2 + \cdots \gamma_d x_d \leq y \leq \gamma_1 x_1 + \cdots + \gamma_d x_d$$